

SIMRAN SAINAND REVANKAR

India | +91 80504 65351 | simirevankar01@gmail.com | LinkedIn | Research website

Research Experience

Project Associate – I | Indian Institute of Technology Palakkad, India | Department of Electrical Engineering | Mar 2025 – Jun 2026

Project: Stochastic Resonance Based Detectors Using Thin-Film Integrated Ferroelectrics

Funding: Core Research Grant, Science and Engineering Research Board (SERB)

Project duration: Dec 2022 – Dec 2025

- Developed reproducible PVDF-TrFE thin-film fabrication workflows using spin coating, Langmuir–Blodgett deposition and drop casting.
- Investigated the influence of solution chemistry, solvent composition, drying and annealing conditions on thin-film properties and electrical behaviour.
- Fabricated ferroelectric capacitor devices and investigated structure-property relationships through electrical measurements.
- Performed P–E hysteresis and PUND measurements using a Radiant Technologies ferroelectric tester.
- Gained exposure to thin-film deposition workflows including sputtering and thermal evaporation within device-fabrication processes.
- Analysed experimental datasets and investigated relationships between processing conditions, thin-film properties and device performance.

Independent Researcher | JSS Science & Technology University, India | Department of Chemistry | Nov 2024 – Mar 2025

Perovskite materials, interfaces and functional polymer systems

- Investigated Cu-based and Rb-based perovskite systems for energy-harvesting applications.
- Examined how composition and nanoscale interfaces influence material and optoelectronic behaviour.

Graduate Researcher | JSS Science & Technology University, India | Department of Chemistry | Dec 2023 – Sep 2024

Halide perovskites, polymer nanocomposites and functional devices

- Synthesized and characterized halide perovskite materials, including lead-based and lead-free compositions, and developed perovskite/polymer nanocomposites for functional devices.
- Investigated relationships between material composition, morphology, interfaces and device performance.
- Fabricated flexible triboelectric devices and evaluated their electrical response under different material compositions and processing conditions.
- Characterized materials using XRD, DSC, SEM, AFM, UV–Vis, FTIR and contact-angle measurements data interpretation.
- Published research on fluorescent perovskite/polymer nanocomposites for self-powered UV sensing.

Research Intern | CSIR–Central Food Technological Research Institute, India | Nov 2023 – Dec 2023

- Optimized curcumin content in oleoresin, extraction was done using spectroscopic and chromatographic analysis.
- Performed characterization using 1D NMR, FTIR, UV–Vis and HPLC.

Research Expertise

Materials and processing: Halide perovskites; lead-based and lead-free compositions; PVDF-TrFE, polyurethane, EVA, PMMA; perovskite/polymer nanocomposites; solution preparation; flexible functional materials.

Characterisation: XRD; FTIR; Raman; contact angle; SEM; AFM; HPLC (data analysis); UV–Vis and 1D NMR (hands-on experience).

Device research: Triboelectric nanogenerators; self-powered UV sensing; electrode/device assembly; electrical-response evaluation.

Research data: Python; OriginPro; ImageJ; Gwyddion; quantitative analysis; scientific visualisation; research documentation.

Publications

- S. S. Revankar, Y. V. Lokesh, A. S. Manjunatha, . . . , B. S. Madhukar, "Fluorescent CsPbBr₃@Cs₄PbBr₆/PU polymer nanocomposite-based triboelectric nanogenerator for self-powered UV sensing," *Materials Today Chemistry*, 44 (2025), 102601. **First author.** DOI: 10.1016/j.mtchem.2025.102601
- N. Manjushree, Y. V. Lokesh, K. R., . . . , S. S. Revankar, . . . , B. S. Madhukar, "Characterization and DFT investigation of CuWO₄@CMC/PVA composite with down conversion ability," *Journal of Molecular Structure*, 1325 (2025), 140780. **7th author.** DOI: 10.1016/j.molstruc.2024.140780
- K. M. Gowda, Y. V. Lokesh, U. G. P. . . . , S. S. Revankar, . . . , B. S. Madhukar, "Flexible, lead-free Cs₃Bi₂Br₉@EVA nanocomposite triboelectric nanogenerator for energy harvesting and tactile sensing," *Emergent Materials*, 8 (2025), 5827–5842. **7th author.** DOI: 10.1007/s42247-025-01170-0

Education

M.Sc. Chemistry | JSS Science & Technology University, India | CGPA: 9.50/10 | GPA: 3.95/4.0 | 2022 – 2024

Thesis: "Flexible and fluorescent CsPbBr₃@Cs₄PbBr₆/PU polymer nanocomposite as triboelectric nanogenerator for self-powered UV sensor."

B.Sc. Chemistry, Zoology & Biotechnology | Maharani Science College for Women, India | CGPA: 8.90/10 | 3.7/4.0 | 2019 – 2022

Research Project: "Survey on PCOS collecting data from 104 individuals."

- Conducted a survey-based study involving data collection from 104 individuals and performed statistical analysis of the collected data.
- Interpreted the findings through discussions with a gynaecologist and a PhD student and prepared the corresponding research report.

Advanced Research Training

- Nanostructured Optoelectronic Materials – NIT Warangal, India, 2025.
- Nanoelectronics & Circuits – IIT Palakkad, India, audited, 2025.

Research Funding & Recognition

- KSCST Student Research Project: "Lead-free Perovskite/Polymer Nanocomposite as Triboelectric Nanogenerator." Funded by the Karnataka State Council for Science and Technology (KSCST), Government of Karnataka; Ref. 47S_MSC_0097; Approved funding: INR 7,000.
- Best Oral Presentation Award, International Conference on Polymers for Sustainable Development, 2025.

Selected Presentations

- Poster Presentation – 17th International Postgraduate Summer School on Green Chemistry, Venice, Italy, 2025.
- Oral Presentation – International Conference on Nanostructured Materials and Nanocomposites, 2024.
- Poster Presentation – International Symposium on Nanomaterials for Health and Energy, 2024.
- Poster Presentation – 47th Student Project Program, State-Level Competition, 2024.

References

Dr. Arvind Ajoy | Associate Professor, Department of Electrical Engineering, Indian Institute of Technology Palakkad, India | +91-62389 06255 | arvindajoy@iitpkd.ac.in

Dr. Madhukar B. S. | Associate Professor, Department of Chemistry, JSS Science & Technology University, India | +91-90369 91340 | madhukarbs007@jssstuniv.in

Dr. Sachin R. Chaudhari | Senior Scientist, Department of Plantation Product, Spices and Flavour Technology, CSIR–Central Food Technological Research Institute, India | +91-96063 50169 | sachinnmr@cftri.res.in